Chapter 10

Hamilton–Jacobi Equations

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 10; printed pp. 543–570; PDF pages 560–587.

Chinese reference: 中译本书内第 384–403 页；扫描 PDF 第 402–421 页。

This chapter reconstructs the source in English with editorial proof details, Chinese mathematical annotations, and native TikZ diagrams. The source's theorem numbers are identified separately from the numbering of these notes. We distinguish the initial-value viscosity convention from the terminal-control convention, and state the extra comparison argument needed for a superlinear Hamiltonian. This is an editorial study draft; it has not undergone independent mathematical review.

10.1 Introduction, Viscosity Solutions

10.1.1 Touching replaces integration by parts

We study

$$u_t + H(D_x u, x) = 0, \qquad u(x,0) = g(x), \tag{10.1}$$

where $H : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ is continuous. Characteristics may intersect, so a globally smooth solution is generally unavailable. The nonlinear expression $H(Du, x)$ has no general distributional interpretation for merely continuous u. At a local extremum of $u - \phi$, however, the derivative of a smooth test function can replace that of u. This is a pointwise contact argument, not integration by parts. Write $\mathrm{BUC}(\mathbb{R}^n \times [0,T])$ for bounded uniformly continuous functions on the whole closed slab; the uniformity includes spatial infinity.

Definition 10.1 (Initial-value viscosity convention; source §10.1.1)
A continuous u is a viscosity subsolution in $\mathbb{R}^n \times (0,T)$ if, whenever $\phi \in C^\infty$ and $u - \phi$ has a local maximum at (x,t),

$$\phi_t(x,t) + H(D\phi(x,t),x) \le 0.$$

A supersolution uses a local minimum and the reverse inequality. A solution is both. For the source's initial-value class we additionally require $u \in \mathrm{BUC}$ on each finite slab and $u(\cdot,0) = g$. The contact need not be strict and its value need not be zero, since a constant can be added to ϕ. All contacts in this definition have $t > 0$.

Lemma 10.2 (Persistence of strict contacts; editorial detail for §10.1.1)
Suppose $u_j \to u$ uniformly on a closed ball B in space–time, $u - \phi$ has a strict maximum at its center z_0, and B lies in the open PDE domain. Maximizers z_j of $u_j - \phi$ on B lie in its interior eventually and converge to z_0.

Expanded proof (the boundary gap and all cluster points).

Continuity and compactness give a positive gap between the central value and the maximum over ∂B. Uniform convergence preserves half that gap, so $z_j \notin \partial B$. Any convergent subsequence of z_j has a limit maximizing $u - \phi$ on B, by uniform convergence. Strictness makes this limit z_0. Compactness then gives convergence of the entire sequence. The minimum version follows by changing signs. For a nonstrict maximum, replace ϕ by $\phi + \delta|z - z_0|^2$: the contact is strict and the first derivatives at z_0 are unchanged. For second-order equations one can instead use $\delta|z - z_0|^4$ to preserve the Hessian as well. □

Theorem 10.3 (Vanishing viscosity; source motivation, made conditional)
Let $\varepsilon_j \downarrow 0$ and let smooth u_j satisfy $(u_j)_t + H(Du_j, x) - \varepsilon_j \Delta u_j = 0$. If $u_j \to u$ locally uniformly, then u solves the first-order equation in the viscosity sense in the open domain. If the convergence includes $t = 0$, the initial values converge to g, and u is in the BUC class, then it is a viscosity solution of the initial-value problem.

Expanded proof (the sign of the positive diffusion).

At a strict upper contact, choose $z_j \to z_0$ by the lemma. There $Du_j = D\phi$, $(u_j)_t = \phi_t$, and $D_x^2 u_j \le D_x^2 \phi$ as symmetric matrices. Hence

$$\phi_t(z_j) + H(D\phi(z_j), x_j) = \varepsilon_j \Delta u_j(z_j) \le \varepsilon_j \Delta \phi(z_j).$$

测试方向与极值

局部最大点对应测试函数从上接触，要求残差不大于零；局部最小点从下接触，要求残差不小于零。导数放在测试函数上，定义不要求 Du 处处存在。

The right side tends to zero because the test Hessian is bounded near z_0. Continuity of H gives the subsolution inequality. At a lower contact the Hessian order reverses and gives the supersolution inequality. Strictification treats arbitrary contacts. Convergence at $t = 0$ gives the trace. Local uniform convergence by itself does not prove global boundedness or uniform continuity, and this theorem assumes the smooth approximants; it does not construct them or establish their uniform estimates. □

10.1.2 Smooth tests, differentiability, and consistency

Lemma 10.4 (C^1 tests suffice; editorial expansion of source Theorem 1)
The first-order definition is unchanged if C^∞ test functions are replaced by C^1 test functions.

Expanded proof (mollify the test, not the unknown).

Only one implication needs proof. Strictify a C^1 upper contact, choose a ball compactly inside the PDE domain, and mollify an extension of ϕ on a slightly larger ball. Then $\phi_j \to \phi$ in C^1 on the contact ball. Maxima of $u - \phi_j$ are interior and converge to the contact point by the boundary-gap argument. Apply the smooth-test inequality and pass to the limit in the first derivatives and H. For a lower contact repeat with minima. A smooth cutoff extends each local test to the ambient domain without changing it near the contact. □

Lemma 10.5 (A differentiable function has a C^1 touching function; source §10.1.2 lemma)
If continuous u is differentiable at z_0, it has a C^1 strict upper test and a C^1 strict lower test there, both with derivative $Du(z_0)$.

Expanded proof (a radial majorant of the differentiability error).

Subtract the tangent affine function and translate so $z_0 = 0$, $u(0) = 0$, $Du(0) = 0$. On a small ball write $|u(z)| \le |z|\rho(|z|)$, where ρ is continuous, nondecreasing and $\rho(0) = 0$. One choice is the maximum over $|w| \le r$ of $|u(w)|/|w|$, extending that quotient by zero at $w = 0$. Set for $r = |z|$

$$\psi(z) = \int_r^{2r} \rho(s)\, ds + r^2.$$

Then $\psi(z) \ge r\rho(r) + r^2 > u(z)$ for $z \ne 0$. For $z \ne 0$ its radial derivative is $2\rho(2r) - \rho(r) + 2r \to 0$, while $\psi(z)/|z| \to 0$. Thus ψ extends as a C^1 function with $D\psi(0) = 0$. It is a strict upper test. The same construction applied to $-u$ gives a lower test for u. Add back the affine function and use a cutoff if a test on the whole ambient domain is desired. □

消黏法的条件式结论

不必控制 D^2u_j 本身：在接触点只需用 $D^2\phi$ 的单侧界。但要得到存在性，还须另证正则化方程可解及紧性，不能由这个极限论证直接宣布一般 H 的存在性。

Theorem 10.6 (Consistency; source Theorem 1)
A C^1 classical solution is a viscosity solution locally. Conversely, a viscosity solution differentiable at (x_0, t_0) satisfies $u_t + H(Du, x_0) = 0$ at that point. A locally Lipschitz viscosity solution therefore solves the equation a.e.

Expanded proof (two contacts recover equality).

For a classical solution, derivatives of u and its test agree at any interior contact, so both test inequalities follow with equality. Conversely, apply the preceding lemma in all $n + 1$ variables. The C^1 test equivalence gives the PDE residual at most zero from the upper test, and at least zero from the lower test. Their derivatives are those of u, proving equality. Rademacher's theorem supplies a.e. differentiability of a locally Lipschitz function. The converse implication from a.e. solvability to viscosity solvability is false: corners carry additional information, as the next diagram and Problem 5 demonstrate. □

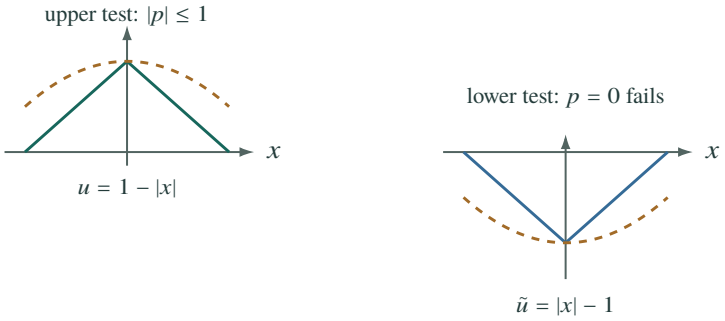


Diagram T10.1. A corner distinguishes a.e. and viscosity solutions. On $(-1, 1)$ both $1 - |x|$ and $|x| - 1$ solve $|u'| = 1$ away from zero and vanish at the endpoints. The upper test $1 - x^2/2$ is allowed for the downward corner, with slope zero and residual $-1 \le 0$. For the upward corner, the lower test $-1 + x^2/2$ has the same residual and violates the required supersolution inequality. No smooth lower test exists at the downward corner.

几乎处处满足还不够

尖点集合可以只有一个点、测度为零，却决定可接受的解。黏性定义在尖点也检查所有可能接触斜率；只在可微点代入方程会漏掉这个选择机制。

10.2 Uniqueness

10.2.1 Tests at the upper time boundary

Lemma 10.7 (Terminal extrema for an initial-value solution; source §10.2 lemma)
Let u be a continuous viscosity subsolution on $0 < t < T$. If $u - \phi$ has a local maximum at (x_0, T) relative to $t \le T$, then $\phi_t + H(D\phi, x_0) \le 0$ there. A supersolution has the reverse inequality at a relative local minimum.

Expanded proof (discard a positive penalty derivative).

Strictify the maximum and restrict to a small closed backward cylinder. For $t < T$ set $\phi_\delta = \phi + \delta/(T - t)$. The quantity $u - \phi_\delta$ tends to $-\infty$ as $t \uparrow T$. A comparison at $(x_0, T - \sqrt{\delta})$ shows its maximum tends to $(u - \phi)(x_0, T)$. The strict gap on the other cylinder boundaries therefore puts maximizing points (x_δ, t_δ) in the open PDE domain, and all cluster points equal (x_0, T). At those points,

$$\phi_t + \frac{\delta}{(T - t_\delta)^2} + H(D\phi, x_\delta) \le 0.$$

Discard the nonnegative middle term and pass to the limit. It need not tend to zero. For a lower contact use $\phi_\delta = \phi - \delta/(T - t)$, and discard its nonpositive derivative from the reverse inequality. This lemma does not reverse the viscosity convention: it only extends the initial-value convention to a one-sided contact at the top of its slab. □

10.2.2 Comparison by doubling space and time

Assume H satisfies, with a single constant C,

$$\begin{aligned} |H(p, x) - H(q, x)| &\le C|p - q|, \\ |H(p, x) - H(p, y)| &\le C|x - y|(1 + |p|). \end{aligned} \tag{10.2}$$

Theorem 10.8 (Comparison; source Theorem 1, strengthened to sub/supersolutions)
Let $u, v \in \mathrm{BUC}(\mathbb{R}^n \times [0, T])$ be a subsolution and a supersolution respectively, with the initial-value convention. If $u(x, 0) \le v(x, 0)$, then $u \le v$ on the slab. In particular there is at most one BUC viscosity solution with prescribed initial data. More generally,

$$\sup_x (u(x, t) - v(x, t)) \le \sup_x (u(x, 0) - v(x, 0)).$$

终端接触与终值问题不同

本引理研究初值解在 $t = T$ 的单侧接触，不改变不等式方向。下一节的终值控制问题采用反向时间演化，才会改变方向；这两个“终端”不能混用。

Expanded proof (localization, small penalty, and two touching tests).

1. A positive maximum after penalization. Suppose $u(x_*, t_*) - v(x_*, t_*) = a > 0$. Fix $\eta > 0$ with $2\eta T < a/4$. For $\varepsilon > 0$ maximize on $\mathbb{R}^{2n} \times [0, T]^2$

$$\begin{aligned}\Phi_\varepsilon(x, y, t, s) = u(x, t) - v(y, s) - \eta(t + s) \\ - \frac{|x - y|^2 + |t - s|^2}{2\varepsilon} - \varepsilon(|x|^2 + |y|^2).\end{aligned} \tag{10.3}$$

Boundedness and the quadratic confinement make the maximum attained; call its point $(x_\varepsilon, y_\varepsilon, t_\varepsilon, s_\varepsilon)$. For small ε, testing on (x_*, x_*, t_*, t_*) shows that this maximum is at least $a/2$. Below all variables refer to this point.

2. First bounds. Comparing with a fixed point, and using boundedness of u, v, gives

$$r := |x - y| = O(\sqrt{\varepsilon}), \quad d := |t - s| = O(\sqrt{\varepsilon}), \quad \varepsilon(|x| + |y|) = O(\sqrt{\varepsilon}).$$

These bounds allow x, y to escape to infinity. This is why uniform continuity on the full slab, rather than just on compact sets, is used.

3. The stronger small-penalty estimate. Compare the maximum with $\Phi_\varepsilon(x, x, t, t)$. If ω_v is a modulus for v, then

$$\frac{r^2 + d^2}{2\varepsilon} \le \omega_v(r + d) + \eta d + \varepsilon(|x| + |y|)r = o(1).$$

In particular $r^2/\varepsilon \to 0$. Merely knowing $r = O(\sqrt{\varepsilon})$ would only bound this expression and would not complete the proof.

4. Neither time is initial. Since the penalty is nonnegative, $u(x, t) - v(y, s) \ge a/2$. If a subsequence had $t \to 0$, then $s \to 0$ as well, and initial ordering plus uniform continuity would give

$$u(x, t) - v(y, s) \le \omega_u(t) + \omega_v(r + s) \longrightarrow 0.$$

This is a contradiction. The same reasoning excludes $s \to 0$. Thus both contact times are positive for all small ε. They may equal T, in which case the preceding lemma applies.

5. Freeze the other pair of variables. At the maximum, the penalty with (y, s) frozen is an upper test for u after adding a constant. Its time derivative and spatial gradient are

$$b_u = \eta + \frac{t - s}{\varepsilon}, \qquad p_u = \frac{x - y}{\varepsilon} + 2\varepsilon x.$$

The negative penalty with (x, t) frozen is a lower test for v, with

$$b_v = -\eta + \frac{t - s}{\varepsilon}, \qquad p_v = \frac{x - y}{\varepsilon} - 2\varepsilon y.$$

Consequently $b_u + H(p_u, x) \le 0$ and $b_v + H(p_v, y) \ge 0$. Subtracting, the uncontrolled time difference cancels:

$$2\eta \le H(p_v, y) - H(p_u, x).$$

6. Estimate the Hamiltonians. By (10.2),

$$\begin{aligned} 2\eta &\le C|p_u - p_v| + Cr(1 + |p_v|) \\ &\le C\sqrt{\varepsilon} + Cr + C\frac{r^2}{\varepsilon} + Cr\sqrt{\varepsilon} \longrightarrow 0. \end{aligned}$$

This contradicts the fixed $\eta > 0$, proving $u \le v$. Adding a constant to v leaves its supersolution property unchanged; apply ordering to $v + \sup_x(u(x,0) - v(x,0))$ for the last assertion. Applying the result in both directions yields uniqueness and sup-norm contraction for solutions with different initial values. □

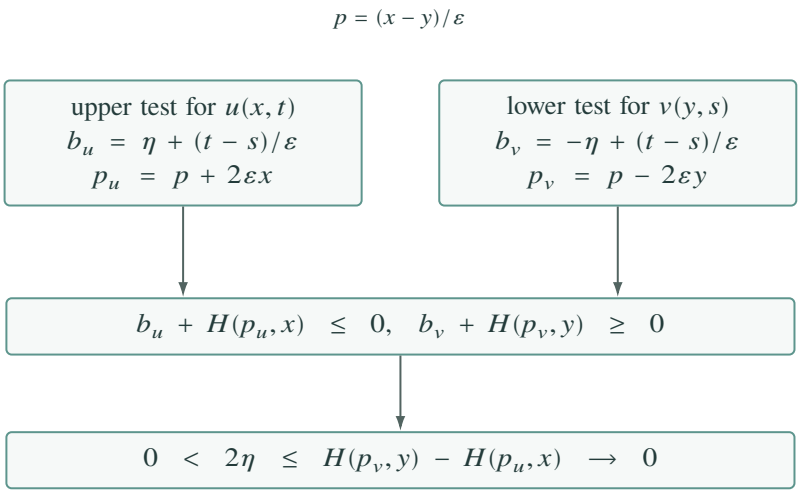


Diagram T10.2. Doubling variables produces two independent smooth tests. The common vector $(x - y)/\varepsilon$ need not be bounded. Global Lipschitz control in momentum handles the small difference $p_u - p_v$, while $|x - y|^2/\varepsilon \to 0$ handles the spatial dependence of H. The opposite strict time shifts leave the positive gap 2η after subtraction.

小量必须足够强

若只得到 $|x-y| = O(\sqrt{\varepsilon})$，则 $|x - y|^2/\varepsilon$ 可能不趋零。对角比较给出整个罚项 $o(1)$，正好消除 Hamiltonian 的空间误差。

10.2.3 A comparison extension for Hopf–Lax

The quadratic Hamiltonian $H(p) = |p|^2/2$ is not globally Lipschitz. The source's Hopf–Lax uniqueness statement therefore needs a separate comparison observation, supplied here.

Theorem 10.9 (One spatially Lipschitz competitor; editorial supplement) *Suppose $H(p)$ is continuous and independent of x. The comparison conclusion above still holds if at least one of u, v is uniformly Lipschitz in x on the slab. The other function may be merely BUC.*

Expanded proof (a contact bounds the momentum).

If a K-Lipschitz function has a differentiable upper or lower touching test at a point, its spatial test gradient has norm at most K. Indeed restrict to increments he, use the touching inequality and the Lipschitz bound, divide by h with both signs, and conclude $|D\phi \cdot e| \le K$ for all unit e.

Use the same doubled maximum and first four steps above. If u is Lipschitz then $|p_u| \le K$; if v is Lipschitz then $|p_v| \le K$. Since $|p_u - p_v| = O(\sqrt{\varepsilon})$, both momenta lie in a fixed compact ball. Uniform continuity of H on that ball gives $H(p_v) - H(p_u) \to 0$, contradicting $2\eta > 0$ exactly as before. No global Lipschitz bound on H, convexity, or boundedness of the other function's gradient is needed for this particular extension. □

超线性 Hamiltonian 的衔接

$H(p) = |p|^2/2$ 的梯度无界，不能直接套用全局动量 Lipschitz 定理。若一方解在空间中为 K-Lipschitz，接触斜率被限制在有界球内，就只需该球上的一致连续性。

10.3 Control Theory, Dynamic Programming

10.3.1 Measurable controls and deterministic responses

Let $A \subset \mathbb{R}^m$ be nonempty and compact. Suppose $f : \mathbb{R}^n \times A \to \mathbb{R}^n$ and $r : \mathbb{R}^n \times A \to \mathbb{R}$ are continuous, and uniformly in $a \in A$ satisfy

$$|f(x,a)| \le M, \quad |f(x,a) - f(y,a)| \le L_f|x-y|,$$
$$|r(x,a)| \le R, \quad |r(x,a) - r(y,a)| \le L_r|x-y|.$$

Let g be bounded and G-Lipschitz. Fix a horizon $T > 0$. An admissible control $\alpha : [t,T] \to A$ is measurable. Its state is an absolutely continuous solution of

$$X'(s) = f(X(s), \alpha(s)) \text{ a.e.}, \qquad X(t) = x. \tag{10.4}$$

Lemma 10.10 (Response estimates; details for source §10.3.1)
Every admissible control has a unique response, and

$$|X(s) - x| \le M(s-t), \qquad |X^{x,\alpha}(s) - X^{y,\alpha}(s)| \le e^{L_f(s-t)}|x-y|.$$

Expanded proof (the integral equation accepts measurable controls).

For fixed α use the map $Z \mapsto x + \int_t^s f(Z(q), \alpha(q))dq$ on continuous paths. The integrand is measurable and bounded; the map returns continuous paths and is a contraction on intervals with $L_f h < 1$. Iterating these intervals gives an absolutely continuous solution on $[t, T]$. Its displacement bound follows by integrating $|f| \le M$. For two responses with the same control, subtract the integral equations and apply Gronwall. This gives the second estimate and uniqueness. No differentiability in time of the control is required. □

The cost and value are

$$\begin{aligned} J_{x,t}(\alpha) &= \int_t^T r(X(s), \alpha(s))ds + g(X(T)), \\ V(x,t) &= \inf_{\alpha} J_{x,t}(\alpha). \end{aligned} \tag{10.5}$$

An infimum does not itself assert existence of an optimal control. The following arguments use controls within an arbitrarily prescribed positive error of the infimum.

10.3.2 The dynamic programming principle

Theorem 10.11 (DPP; source §10.3.2, Theorem 1)
For $0 < h \le T - t$,

$$V(x,t) = \inf_{\alpha|_{[t,t+h]}} \left\{ \int_t^{t+h} r(X(s), \alpha(s))ds + V(X(t+h), t+h) \right\}. \tag{10.6}$$

Expanded proof (restriction and concatenation give opposite bounds).

Call the right side W. Any full control has a prefix up to $t + h$ and a continuation thereafter. Its continuation cost is at least the value at its reached state. Thus every full cost is at least W, giving $V \ge W$.

Conversely, fix any prefix, with reached state z. For $\delta > 0$ choose a continuation whose cost is at most $V(z, t+h) + \delta$. Concatenate the two measurable controls. Uniqueness of the state equation makes its trajectory exactly the concatenation of their responses. Therefore $V(x,t)$ is at most the displayed prefix cost plus δ. Take the infimum over prefixes and let $\delta \downarrow 0$, obtaining $V \le W$. A continuation is chosen for a single reached

紧控制集保证什么

连续性与紧性保证 Hamiltonian 的逐点最小值能取到。但这并不自动给出整个时间区间的最优控制，也不自动保证反馈场的闭环 ODE 可解；动态规划证明只使用近似最优控制。

state at a time; no measurable selection depending simultaneously on every state is needed. □

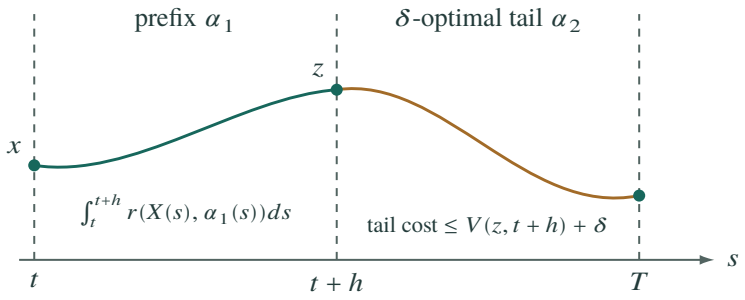


Diagram T10.3. A prefix and its continuation form one admissible control. Restricting a full control gives a continuation cost at least $V(z, t+h)$. Conversely, for a fixed prefix reaching z, choose one δ-optimal continuation and concatenate it. The two comparisons prove the equality in DPP. The curves are schematic state trajectories, not a claim of optimality for the drawn paths.

Lemma 10.12 (Regularity of the value; source §10.3.3 lemma)
The value is bounded by $RT + \|g\|_\infty$. Uniformly in t it is K_T-Lipschitz in x, where

$$K_T = Ge^{L_f T} + L_r \int_0^T e^{L_f s} ds.$$

It is $(R + K_T M)$-Lipschitz in time and $V(x, T) = g(x)$.

Expanded proof (compare identical controls before taking infima).

For the same control started at x, y, the response estimate bounds the cost difference by

$$\left(Ge^{L_f (T-t)} + L_r \int_0^{T-t} e^{L_f s} ds \right) |x - y|.$$

This holds for every control, so taking infima yields the same bound for $|V(x,t) - V(y,t)|$. In DPP every reached state satisfies $|X(t+h) - x| \le Mh$ and every running integral has absolute value at most Rh. The expression inside its infimum is therefore between $V(x, t+h) - (R + K_T M)h$ and $V(x, t+h) + (R + K_T M)h$. Taking its infimum gives the time estimate without assuming a smooth value function or an optimal control. At $t = T$ the running interval is empty and $X(T) = x$, so every control gives cost $g(x)$. □

10.3.3 The terminal HJB equation and its time orientation

Define the control Hamiltonian

$$H_c(p,x) = \min_{a\in A}\{f(x,a)\cdot p + r(x,a)\}. \tag{10.7}$$

It is concave in p, since it is a minimum of affine functions, and

$$|H_c(p,x) - H_c(q,x)| \le M|p-q|,$$
$$|H_c(p,x) - H_c(p,y)| \le (L_f|p| + L_r)|x-y|.$$

For a terminal problem written as $V_t + H_c(DV,x) = 0$, the source uses the following convention: an upper contact has residual $\phi_t + H_c(D\phi,x) \ge 0$, and a lower contact has residual at most zero. Equivalently this is the usual viscosity convention applied to $-V_t - H_c(DV,x) = 0$.

Lemma 10.13 (Time reversal; source remark after Theorem 2)
Set $w(x,\tau) = V(x,T-\tau)$. Then the terminal convention for V is exactly the initial-value convention for

$$w_\tau + K(Dw,x) = 0, \qquad K(p,x) = -H_c(p,x), \qquad w(x,0) = g(x).$$

In particular K is convex in p and satisfies (10.2).

Expanded proof (transform the test as well as the equation).

If $w-\psi$ has an upper contact at (x_0,τ_0), then $V-\phi$, $\phi(x,t) = \psi(x,T-t)$, has an upper contact at $(x_0,T-\tau_0)$. Now $\phi_t = -\psi_\tau$, $D\phi = D\psi$. The terminal inequality $-\psi_\tau + H_c(D\psi,x) \ge 0$ is precisely $\psi_\tau - H_c(D\psi,x) \le 0$. Lower contacts give the other direction. The implication is reversible. Negating the minimum of affine functions makes a maximum of affine functions, proving convexity of K. □

先用同一控制比较

估计两个初值的价值差时，对它们使用相同控制，先用 Gronwall 控制轨道差，再对代价取下确界。直接把两个未知最优控制相减没有这样的稳定性依据。

Theorem 10.14 (Value function solves HJB; source Theorem 2)
The value in (10.5) *is the unique BUC solution of* $V_t + H_c(DV, x) = 0$, $V(x,T) = g(x)$, *with the terminal convention just specified. Equivalently its time reversal solves the corresponding initial-value problem with Hamiltonian* $K = -H_c$.

Expanded proof (all constant controls versus an h^2-optimal prefix).

Normalize an upper test by $V(x_0,t_0) = \phi(x_0,t_0)$. For any fixed $a \in A$, use the constant control on $[t_0, t_0 + h]$. The displacement bound keeps its path in the contact neighborhood when h is small, independently of a. DPP and upper touching give

$$\begin{aligned} 0 &\le \int_{t_0}^{t_0+h} r(X(s),a)ds + V(X(t_0+h), t_0+h) - V(x_0,t_0) \\ &\le \int_{t_0}^{t_0+h} (\phi_t + f(X(s),a)\cdot D\phi + r(X(s),a))ds. \end{aligned}$$

Divide by h and let $h \downarrow 0$. The limit inequality holds for every a, hence $\phi_t(x_0,t_0) + H_c(D\phi(x_0,t_0),x_0) \ge 0$.

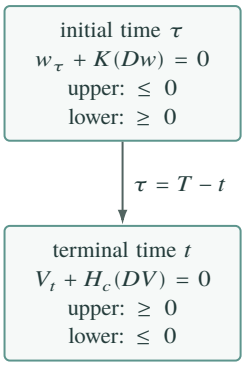
At a normalized lower contact choose a prefix within h^2 of the DPP infimum. The lower touching inequality now yields

$$h^2 \ge \int_{t_0}^{t_0+h} (\phi_t + f(X(s),\alpha(s))\cdot D\phi + r(X(s),\alpha(s)))ds.$$

Continuity in (x,t,a) on a compact neighborhood times A makes replacement of $(X(s),s)$ by (x_0,t_0) have a uniform $o(h)$ error. The resulting integrand is bounded below by $\phi_t(x_0,t_0) + H_c(D\phi(x_0,t_0),x_0)$ for every control value. After division by h and passage to the limit we obtain a residual at most zero. The error must be $o(h)$: a fixed δ-optimal control would leave the uncontrolled quotient δ/h.

The value is BUC and takes the terminal data by the regularity lemma. Time reversal converts it to an initial-value solution for K, whose structural estimates allow comparison in §10.2. This proves uniqueness. □

图 T10.M：时间方向与测试符号



图中残差均按方程所写的时间变量计算。令 $\tau = T - t$ 后，$\partial_t = -\partial_\tau$，Hamiltonian 同时变为 $-H_c$；不能只改变时间端点而保留所有符号。

近似最优误差要随步长缩小

在下接触证明中取 h^2-最优前缀，再除以 h，误差才趋零。上接触使用每个常值控制；下接触使用近似最优控制，二者的量词不同。

10.3.4 Verification and the limits of feedback synthesis

Theorem 10.15 (Smooth verification; editorial completion of source discussion) *Suppose $v \in C^1(\mathbb{R}^n \times [0,T])$ satisfies $v_t + H_c(Dv, x) = 0$ and $v(x,T) = g(x)$. Then $v(x,t) \le J_{x,t}(\alpha)$ for every admissible control. If an admissible α_* with response X_* attains the Hamiltonian minimum along its path a.e., then its cost equals $v(x,t)$ and it is optimal; in particular $v = V$ at that starting point.*

Expanded proof (integrate the nonnegative verification residual).

The chain rule for an absolutely continuous path gives

$$\frac{d}{ds}v(X(s),s)+r(X(s),\alpha(s)) = v_t+f(X(s),\alpha(s))\cdot Dv+r(X(s),\alpha(s)) \ge 0$$

a.e. Integrate to T and use the terminal value g to obtain $J_{x,t}(\alpha) \ge v(x,t)$. If the minimum is attained along X_*, the residual vanishes a.e. and equality holds. All evaluations are along a bounded path on a finite interval, where the derivatives of v are bounded. This verifies an admissible path; it does not establish solvability of a closed-loop ODE for an arbitrary minimizing feedback field. □

The pointwise prescription $a_*(x,t) \in \operatorname{argmin}_{a\in A}\{f(x,a)\cdot DV(x,t) + r(x,a)\}$ is conditional on the gradient and on an admissible resulting trajectory. At nonsmooth points DV may not exist. Even with a pointwise minimizer, measurability and closed-loop existence require additional arguments.

10.3.5 Hopf–Lax: compact minimizers and viscosity inequalities

Return now to an initial-value Hamiltonian $H : \mathbb{R}^n \to \mathbb{R}$, finite, convex and superlinear, with g bounded and G-Lipschitz. Let

$$L(q) = \sup_p\{p\cdot q - H(p)\}, \qquad Q_t g(x) = \min_y \left\{tL\left(\frac{x-y}{t}\right) + g(y)\right\}.$$

Convex duality from §3.3 gives finite continuous convex superlinear L and $H(p) = \sup_q(p\cdot q - L(q))$. In particular all displayed minima exist by coercivity. These assumptions include $H(p) = |p|^2/2$, which is outside the global momentum Lipschitz hypothesis of §10.2.

验证定理与控制构造

验证定理说：若已有合适的最小化轨道，则它最优。它没有证明任意逐点选出的反馈都可积分成轨道；尤其在价值函数不可微处，直接写 DV 没有定义。

Lemma 10.16 (Semigroup, speed bound, and regularity; details for source Theorem 3)
One has $Q_h Q_t g = Q_{h+t} g$. Each $Q_t g$ is G-Lipschitz in space. All minimizing velocities $q = (x-y)/t$ lie in a ball independent of x, t. With $Q_0 g = g$, the function $U(x,t) = Q_t g(x)$ is BUC on every finite slab and is Lipschitz in time.

Expanded proof (Jensen and the competitor with zero velocity).

For a two-stage path from y through z to x, Jensen gives

$$tL\left(\frac{z-y}{t}\right) + hL\left(\frac{x-z}{h}\right) \geq (t+h)L\left(\frac{x-y}{t+h}\right).$$

Equality is obtained by $z = (tx+hy)/(t+h)$, so infimizing proves the semigroup identity. Translating x and y by the same increment changes only $g(y)$, proving the spatial Lipschitz estimate.

At a minimizer compare with $y = x$. Since g is G-Lipschitz, $L(q) - G|q| \leq L(0)$. Superlinearity makes this sublevel bounded, say $|q| \leq R_G$. The bound is uniform even as $t \downarrow 0$. For any G-Lipschitz v, set $m_G = \inf_q (L(q) - G|q|) > -\infty$. Then

$$v(x) + hm_G \leq Q_h v(x) \leq v(x) + hL(0).$$

Together with the semigroup this gives a uniform time Lipschitz constant $\max\{|m_G|, |L(0)|\}$ and convergence to g at $t = 0$. The same estimate with $v = g$ bounds U on finite slabs. □

Theorem 10.17 (Hopf–Lax viscosity solution; source §10.3.4, Theorem 3)
Under the preceding assumptions $U(x,t) = Q_t g(x)$ is the unique BUC viscosity solution of $U_t + H(DU) = 0$, $U(x,0) = g(x)$.

Expanded proof (bounded minimizing velocities close the lower test).

At an upper contact at (x,t), choose any fixed velocity q and $0 < h < t$. The semigroup gives $U(x,t) \leq hL(q) + U(x-hq, t-h)$. Upper touching then gives $\phi(x,t) - \phi(x-hq, t-h) \leq hL(q)$. Divide by h and let $h \downarrow 0$ to get $\phi_t + D\phi \cdot q - L(q) \leq 0$ for every q. Taking the supremum in q yields $\phi_t + H(D\phi) \leq 0$.

At a lower contact, let q_h minimize the short-step semigroup formula, so $U(x,t) = hL(q_h) + U(x - hq_h, t-h)$. The spatial Lipschitz estimate for $U(\cdot, t-h)$ bounds $|q_h| \leq R_G$ by the preceding lemma. Thus the earlier point stays in the contact neighborhood and Taylor's first-order remainder is $o(h)$ uniformly in q_h. Lower touching gives

$$hL(q_h) \leq \phi(x,t) - \phi(x - hq_h, t-h) = h(\phi_t + D\phi \cdot q_h) + o(h).$$

Since $D\phi \cdot q_h - L(q_h) \le H(D\phi)$, $\phi_t + H(D\phi) \ge -o(1)$, which proves the lower-test inequality. Initial data and BUC regularity were proved in the lemma.

For uniqueness compare U with any other BUC solution in both directions. The extension at the end of §10.2 applies because U is uniformly spatially Lipschitz and H is continuous and independent of x. This explicitly supplies the comparison step without assuming a global Lipschitz bound for the superlinear H. □

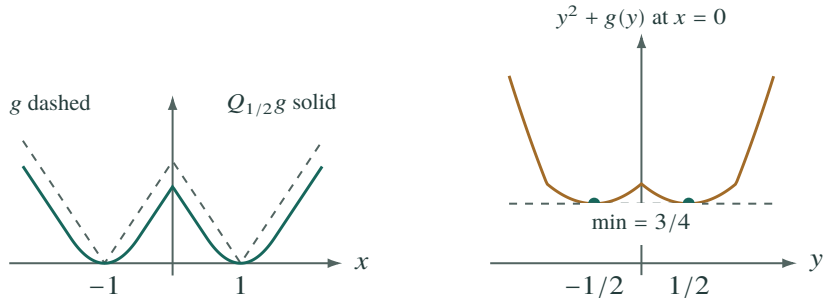


Diagram T10.4. Two minimizing origins can create a viscosity corner. Take $H(p) = p^2/2$, $t = 1/2$, and $g(y) = \min\{|y-1|, |y+1|, 2\}$. The exact value is $\min\{h_t(|x-1|), h_t(|x+1|), 2\}$, where $h_t(d) = d^2/(2t)$ for $d \le t$ and $d - t/2$ otherwise. At $x = 0$ the two minimizing origins are $y = \pm 1/2$, both of cost $3/4$. The left panel shows g and its value; the right panel shows the minimized cost at $x = 0$.

10.4 Problems

The descriptions below retain the source numbers 1–14. All solutions are editorial reconstructions. The viscosity convention is stated whenever the time direction or the sign of the equation changes.

1. Negating the unknown changes the Hamiltonian

Solution. Let $w = -u$ and $\widehat{H}(p,x) = -H(-p,x)$. If $w - \phi$ has an upper contact, $u - (-\phi)$ has a lower contact. The supersolution inequality for u reads $-\phi_t + H(-D\phi, x) \ge 0$, exactly $\phi_t + \widehat{H}(D\phi, x) \le 0$. Lower contacts reverse the reasoning. Boundedness and uniform continuity are preserved, as are initial traces with g replaced by $-g$. Negating the unknown interchanges upper and lower contacts; multiplying a PDE expression by -1 while leaving the unknown and the test inequalities unchanged does not do this. □

最小化速度为何有界

由与 $y = x$ 比较得 $L(q) - G|q| \le L(0)$。超线性把 q 限在固定球内，因此短步最小化点满足 $|x-y| = O(h)$，下接触证明中的 Taylor 余项才一致为 $o(h)$。

2. Uniform stability for a first-order equation

Solution. At a strict upper contact for the limit, maximize $u_k - \phi$ on a small closed contact ball. Uniform convergence puts the maxima in its interior and makes their points converge to the original contact point. Apply $\phi_t + H(D\phi, x) \le 0$ there and use continuity of H. Strictification and lower contacts give the full PDE. A uniform limit on each full closed slab of BUC functions remains BUC and inherits the limiting initial trace. If only local uniform convergence is known, the local PDE conclusion still holds, but membership in the global BUC class must be checked separately. The same contact proof works for $H_k \to H$ locally uniformly in (p, x): the test gradients stay in a compact set even when the unknown functions have no gradient bounds. □

3. Anisotropic positive diffusion selects the same notion

Solution. Let the fixed smooth matrix $A = (a_{ij})$ have nonnegative symmetric part (the source assumes uniform ellipticity). At a smooth upper contact, $D^2 u^\varepsilon \le D^2\phi$, and hence

$$\phi_t + H(D\phi, x) = \varepsilon \operatorname{tr}(AD^2 u^\varepsilon) \le \varepsilon \operatorname{tr}(AD^2\phi).$$

To see the trace order, for positive semidefinite symmetric A, B, $\operatorname{tr}(AB) = \operatorname{tr}(A^{1/2}BA^{1/2}) \ge 0$; the antisymmetric part has zero contraction with a symmetric Hessian. The contact points converge, A and $D^2\phi$ are bounded locally, and the right side tends to zero. Lower contacts yield the reverse inequality. Boundary traces and the BUC class are treated as in Problem 2. Uniform ellipticity helps construct approximants but is stronger than the positivity needed in this limit argument. □

4. Sup-norm contraction

Solution. Set $m = \sup_x |g_1 - g_2|$. Since H contains no zeroth-order value variable, $u_2 + m$ remains a supersolution and $g_1 \le g_2 + m$. Comparison gives $u_1 \le u_2 + m$. Interchange indices to obtain the opposite inequality. Thus $\|u_1(\cdot, t) - u_2(\cdot, t)\|_\infty \le m$ on every finite slab. There is no requirement that either supremum be attained at a finite spatial point. □

5. The sign of the eikonal equation matters

Solution. For $u = 1 - |x|$, every nonzero point has slope ± 1, so the equation holds classically there. If a smooth ϕ touches from above at 0, the right and left one-sided inequalities give $-1 \le \phi'(0) \le 1$. Thus $|\phi'(0)| - 1 \le 0$. There is no smooth lower test at 0: it would require both $\phi'(0) \le -1$ from the right and $\phi'(0) \ge 1$ from the left. Hence the lower-test condition is vacuous there. This proves (a).

For $\tilde{u} = |x| - 1$, the constant lower test $\phi = -1$ at zero violates $|\phi'(0)| - 1 \geq 0$, proving (b). For (c), use instead $F(p) = 1 - |p|$. The lower-test slopes for $\tilde{u}$ lie in $[-1, 1]$, so $F \geq 0$, and it has no upper test at its upward corner. Away from zero, $F = 0$. Thus $\tilde{u}$ solves the sign-reversed equation in its own viscosity sense. For (d), the two residuals have the same classical zero set, but their upper- and lower-test inequalities are different. They encode opposite choices at a corner, exactly as Diagram T10.1 illustrates. □

方程乘负号会改变选择

经典方程 $F(Du) = 0$ 与 $-F(Du) = 0$ 等价，但黏性定义包含残差的方向。若未知函数不变，乘负号会交换应满足的单侧条件，因此尖点解可能不同。

6. Distance to the boundary is an eikonal solution

Solution. For any nonempty bounded open U, $d(x) = \operatorname{dist}(x, \partial U)$ is 1-Lipschitz by the triangle inequality; no boundary smoothness is required. At an upper smooth contact its gradient has norm at most one by the Lipschitz contact estimate proved in §10.2. At a lower contact $x_0 \in U$, choose a nearest boundary point y, possible because ∂U is closed and bounded. Set $e = (y - x_0)/d(x_0)$. For $0 < h < d(x_0)$ the point $x_0 + he$ is inside U, and

$$d(x_0 + he) = d(x_0) - h.$$

The upper bound uses the same y; the lower bound uses the Lipschitz estimate. Lower touching gives $\phi(x_0 + he) - \phi(x_0) \leq -h$, hence $D\phi(x_0) \cdot e \leq -1$. Therefore $|D\phi(x_0)| \geq 1$, proving the supersolution condition. The continuous extension of d vanishes at the boundary. □

7. Admissible gradient jumps at a smooth interface

Solution. Continuity of u and smoothness of its traces force the tangential components of Du^+ and Du^- to agree on Γ. At a fixed point use smooth local coordinates (z, ρ), where ρ increases into V^+ and $\partial_\rho = \nu$ at that point. If $u_\nu^- < u_\nu^+$, choose any number a strictly between them. For a sufficiently small neighborhood the smooth function

$$\phi(z, \rho) = u(z, 0) + a\rho$$

is a lower test along the interface: on $\rho > 0$ the normal derivative of u exceeds a, while on $\rho < 0$ it is less than a. Integrating those derivative inequalities from zero gives $u \geq \phi$ on both sides. Its gradient at the point is the common tangential gradient plus $a\nu$. The viscosity lower-test condition gives $H(\lambda Du^- + (1 - \lambda)Du^+) \geq 0$ for $0 < \lambda < 1$. The endpoint values of H are zero by the classical equation on each side and continuity up to Γ; thus the closed interval follows. If the normal derivatives agree, both full gradients agree and the same endpoint argument suffices. For a downward jump the function ϕ is an upper test, giving $H \leq 0$ on the

segment. This proves both source inequalities, including equality cases. □

界面斜率形成一个线段

连续性固定切向梯度，只有法向分量能跳跃。上凸或下凸尖角的测试斜率填充两侧梯度之间的线段，因此要检查整段上的 H，而非只检查两端的 $H = 0$。

8. Shape from shading

Solution. For the graph $(x_1, x_2, u(x))$, the upward unit normal is $\nu = (-Du, 1)/\sqrt{1 + |Du|^2}$. Hence the normalized Lambertian intensity is $i = e_3 \cdot \nu = (1 + |Du|^2)^{-1/2}$. For $0 < i(x) \le 1$ this gives the eikonal equation

$$|Du| = n(x) := \sqrt{i(x)^{-2} - 1}.$$

If $i = 1$ the slope is zero; if $i \downarrow 0$ the required finite-graph slope diverges. Values outside $(0, 1]$ do not fit this normalized graph model. The PDE determines a slope magnitude, not a surface without boundary data and a solution-selection condition. □

9. Pure pursuit and an exact interception time

Solution. Use relative coordinates x_1 for the target's horizontal displacement from the pursuer and x_2 for the pursuer's height above the target. Writing $r = \sqrt{x_1^2 + x_2^2}$, the relative dynamics before interception are

$$x_1' = b_1 - b_2 x_1/r, \qquad x_2' = -b_2 x_2/r.$$

The remaining-time identity $u(x) = h + u(X(h))$ gives

$$(b_1 - b_2 x_1/r)u_{x_1} - (b_2 x_2/r)u_{x_2} = -1, \qquad u(0,0) = 0.$$

As an editorial check the exact solution is

$$u(x_1, x_2) = \frac{b_2\sqrt{x_1^2 + x_2^2} + b_1 x_1}{b_2^2 - b_1^2}.$$

Indeed $r' = b_1 x_1/r - b_2$, so differentiating the numerator along a trajectory gives $-(b_2^2 - b_1^2)$. Also $r' \le b_1 - b_2 < 0$ ensures finite interception, and the displayed expression tends to zero there. Thus its constant rate of decrease -1 identifies it as the actual time. On $x_2 = 0, x_1 > 0$ it reduces to $x_1/(b_2 - b_1)$, as expected. The transport field is singular at interception; the PDE is used only away from $r = 0$ and the zero-time value is imposed by continuity. □

10. Discount and regularity on an infinite horizon

Solution. For $\lambda > 0$ define $V(x) = \inf_\alpha \int_0^\infty e^{-\lambda s} r(X(s), \alpha(s))ds$. Since $|r| \le R$, $|V| \le R/\lambda$. For the same control at x, y, the running-cost difference is at most $\min\{L_r e^{L_f s}|x - y|, 2R\}$. If $\lambda > L_f$, integrate its first

bound to get $|V(x) - V(y)| \le L_r|x-y|/(\lambda - L_f)$. If $0 < \lambda \le L_f$, choose any $0 < \alpha < \lambda/L_f$ (and hence $\alpha < 1$). For nonnegative a, b, $\min(a,b) \le a^\alpha b^{1-\alpha}$, so

$$|V(x) - V(y)| \le \frac{L_r^\alpha (2R)^{1-\alpha}}{\lambda - \alpha L_f}|x-y|^\alpha.$$

The estimates are uniform in the control, so passing to infima preserves them. The case $R = 0$ or $L_r = 0$ is immediate. If $L_f = 0$, every $\lambda > 0$ is in the Lipschitz case. At $\lambda = L_f$ the stated argument gives every $\alpha < 1$, without asserting Lipschitz continuity. □

11. The discounted stationary Bellman equation

Solution. Restriction and concatenation, with the discount factored out of the tail, give

$$V(x) = \inf_\alpha \left\{ \int_0^h e^{-\lambda s} r(X(s), \alpha(s)) ds + e^{-\lambda h} V(X(h)) \right\}.$$

At a normalized upper contact $V - \phi$, choose a constant control a. The expression on the right for this control is at least $V(x)$; replace $V(X(h))$ by its upper test and expand to first order: $0 \le r(x,a) + f(x,a) \cdot D\phi(x) - \lambda V(x)$. This holds for every a, so

$$\lambda V(x) - \min_a \{ f(x,a) \cdot D\phi(x) + r(x,a) \} \le 0.$$

At a lower contact choose an h^2-optimal prefix, use the lower test, and the uniform continuity in a argument of §10.3. Division by h now gives the reverse inequality. Thus V is a viscosity solution of $\lambda V - H_c(DV, x) = 0$ with the usual stationary convention. The positive coefficient of V is produced by differentiating the discount factor, not by inserting an additional running cost. □

12. Degenerate ellipticity and extrema of operators

Solution. Write $R \ge S$ when $R - S$ is positive semidefinite. Then $-\operatorname{tr} R \le -\operatorname{tr} S$. For symmetric $A \ge 0$, $\operatorname{tr}(A(R-S)) = \operatorname{tr}(A^{1/2}(R-S)A^{1/2}) \ge 0$, so $F(R) = -A : R$ is elliptic in the source's decreasing-matrix convention. If every $F_k(R,p,x) \le F_k(S,p,x)$, taking a finite maximum or a finite minimum preserves that inequality. Consequently both Bellman-type maxima and minima of such linear operators are degenerate elliptic. Strict positive definiteness is not needed for these order statements. □

13. Classical consistency for second-order equations

Solution. Let $u \in C^2$ solve $F(D^2u, Du, x) = 0$ with F elliptic as in Problem 12. At an upper smooth contact, $Du = D\phi$ and $D^2u \le D^2\phi$, so

$$F(D^2\phi, D\phi, x) \le F(D^2u, Du, x) = 0.$$

At a lower contact both the matrix order and the resulting inequality reverse. This proves viscosity consistency. The ellipticity sign is essential: it is exactly what connects Hessian order at a contact to the direction of the PDE inequality. □

14. Uniform stability for second-order equations

Solution. Assume, as in the preceding problem, that F is continuous and elliptic, and that $u_k \to u$ locally uniformly. Strictify an upper contact by adding $\delta|x - x_0|^4$ to ϕ. This preserves both the gradient and Hessian at x_0. Interior maxima of $u_k - \phi_\delta$ converge to x_0. At those points use the viscosity subsolution inequality and pass to the limit in the continuous F, including the derivatives of the fixed test. At x_0 the quartic perturbation contributes no first or second derivative, giving the desired original-test inequality. Strictify a lower contact by subtracting the quartic and repeat. This is a stability proof; it does not prove existence or comparison for general fully nonlinear second-order equations. □

10.5 References

Source attributions

Evans's §10.5 attributes the test-function formulation to Crandall, Evans and Lions, Trans. AMS 282 (1984), 487–502, following the foundational Crandall–Lions paper, Trans. AMS 277 (1983), 1–42. The uniqueness argument follows the first paper, with improvements acknowledged there to M. Crandall. The control connection is attributed to P.-L. Lions [Li]. Further directions in the source are Tran [Tn], Fleming–Soner [F-S], and Bardi–Capuzzo Dolcetta [B-CD]. These are source attributions and bibliography keys. The source credits D. Ostrov with Problem 8 and directs readers to Cabré–Caffarelli [C-C] and Gilbarg–Trudinger [G-T] for the fully nonlinear elliptic exercises. The present reconstruction does not claim separate consultation of all those works.

Dependencies and proof boundaries

The proofs use compactness on contact balls, mollification, elementary convex duality from Chapter 3, Rademacher's theorem, the integral form of the ODE and Gronwall. The comparison proof is first-order: freezing the other variables produces smooth tests directly. No second-order matrix jet lemma is being assumed or proved. The source's uniqueness theorem is expanded to comparison of BUC sub- and supersolutions. An additional comparison theorem with one spatially Lipschitz competitor explicitly supplies the superlinear Hopf–Lax uniqueness step.

The vanishing-viscosity argument is conditional on smooth approximants and locally uniform convergence; existence and uniform estimates for a general nonconvex Hamiltonian are not proved here. Value-function existence is proved for the stated bounded continuous coefficients and compact control set; its terminal Hamiltonian is concave, and the time-reversed initial Hamiltonian is convex. Smooth verification is conditional on an admissible minimizing trajectory. General nonsmooth feedback synthesis, existence of an optimal ordinary control under additional hypotheses, second-order comparison, Perron's method, and nonlocal or state-constrained boundary theories are outside this chapter's completed proofs. The 14 problem solutions and all diagrams are editorial additions. Source checking, symbolic model checking, compilation, and visual QA are recorded separately; none is an independent mathematical review.